Predicting Invasive Asian Carp Movement in the Mississippi River

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Introduction

- Asian Carp were introduced into the U.S. in the early 1970s to control algae in wastewater ponds.
- During the mid-to-late 1970s, floods allowed the carp to escape into the Mississippi River Basin.
- They damage riverbeds, compete with and starve native species, and jump out of the water when startled.
- Continued upstream movement could impact additional aquatic ecosystems.

Abstract

- This project will use predictive models to identify high risk river conditions as well as models that will try and link carp movement to Lock and Dam conditions so that predictions could be made.
- The model demonstrated in this poster is a Random Forest model which combines the predictions from many different, independent decision trees to produce a more accurate and stable final result.
- Key factors for prediction are weight, length, species, week.
- This model does well with predictions overall but sometimes struggles with upstream passage prediction.

Objectives

- Gather information from the U.S Army Corps of Engineers
- Clean and format data in a way that suits prediction
- Use the cleaned data to create predictive models to figure out the conditions in which the carp are most likely to move upstream.

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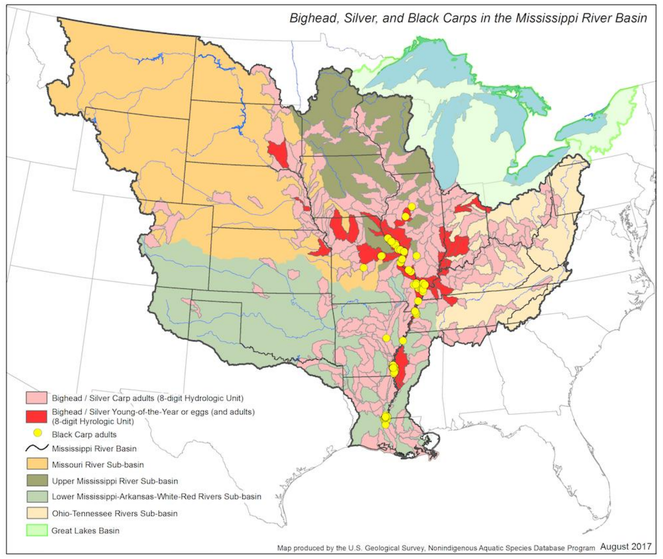
Materials & Methods

- R is the programming language used in project
- This model uses the Random Forest library for the general model
- Other libraries include: ggplot2, pROC, lubridate, dyplr
- Cleaned and preprocessed fish telemetry data in R (handled missing values, factored categories, log-transformed skewed variables).
- A Random Forest model predicted upstream passage using environmental, biological, and lock activity variables.
- Data were split 70/30 by transmitter ID to avoid the same fish appearing in both sets. Balanced sampling addressed the rarity of upstream passage events.
- Evaluated model performance using a confusion matrix, classification metrics (accuracy, precision, recall, F1), identified key predictors by importance.



Conclusions

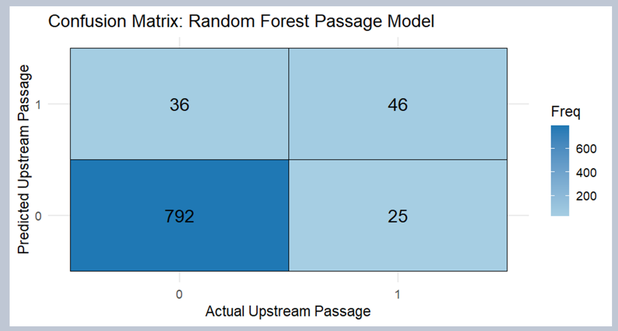
- The Random Forest model accurately predicts non-passage but struggles with rare upstream events, lowering class 1 precision and recall.
- Upstream passage is a low-frequency event, which limits the model's ability to predict these occurrences accurately.
- Future models planned are spatial and risk-based models to better capture movement patterns and upstream passage conditions.
- Improving prediction for rare passage events will require enhanced data representation or targeted sampling to better capture the minority class.

References

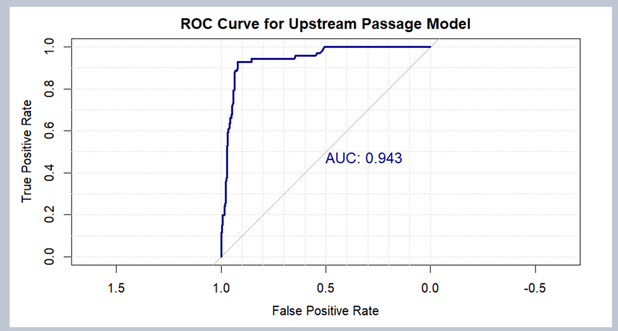
- R Core Team (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.
- RStudio Team (2025). RStudio: Integrated Development Environment for R. Posit Software, PBC, Boston, MA.
- Fritts, A., & Knights, B. (2020, November 30). 2017-2018 telemetry data for Asian carp and native fish species at lock and dam 19 in the upper mississippi river basin. sciencebase.gov. https://www.sciencebase.gov/catalog/file/get/5fbe7c1ad34e4b9faad7e8cc?f=__disk__14%2F2a%2F6b%2F142a6bfa9ab133230a0c0a6e45f2b278541dde1c&transform=1&allowOpen=true
- Robin, X., Turck, N., Hainard, A., Tiberti, N., Lisacek, F., Sanchez, J.-C., & Müller, M. (Year). pROC: Display and analyze ROC curves (R package version X.X.X). CRAN. https://cran.r-project.org/web/packages/pROC/pROC.pdf
- Görög, L. (2022, September 5). R packages for Predictive Analytics. Predictea Digital Care - AI Strategy, Predictions, Data Analysis. https://www.predictea.com/r-packages-for-predictive-analytics/

Results

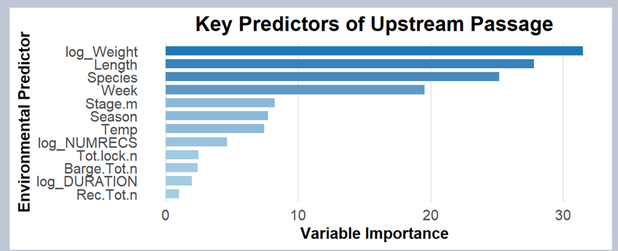
- Overall model accuracy was 93.2%.
  - Precision (0.56): Of all predicted passage events, 56% were correct.
  - Recall (0.65): The model identified 65% of actual passage events.
  - F1 score (0.60): Balances precision and recall for the rare passage events.
- The confusion matrix shows that the model predicts non-passage events very well (792/817 correct) but is less precise for passage events (46/82 correct)



- The ROC curve shows an AUC of 0.943, indicating excellent ability to discriminate between passage and non-passage events.



- Variable importance shows that fish weight, species, length, and week of year are the strongest predictors of upstream passage.



- Upstream passage is most likely during the spring and summer when water temperatures are up.
- The invasive species of carp most like to move upstream during these conditions is the bighead carp.

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